

Maria Polak DSP Computer Scientist

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Hi, my name is Maria, and I am a computer scientist with a passion for signal processing, a music geek and an amateur puzzle-solver. During my experiences in real-time audio software, machine learning, and full-stack development I have found myself being passionate about solving complex algorithmic and technical challenges. I consider myself a cheerful, detail-oriented, and hardworking person.

PROFESSIONAL EXPERIENCE

Audio Software Engineer & Owner, Spectral Plugins FEB 2021 – SEP 2023

2 YEARS 7 MONTHS • CO-OWNER • CRACOW, POLAND

REAL-TIME DEVELOPMENT C++ DSP AZURE PIPELINES BUSINESS INTELLIGENCE

Developed and maintained cross-platform audio plugins using C++17 and JUCE framework. Designed real-time audio effects based on DSP techniques. Maintained self-owned business, including website, servers and marketing. Successfully released 3 industry-recognized audio plugins.

C++ Software Developer, Ericsson SEP 2022 - MAY 2023

9 MONTHS • FULL-TIME • CRACOW, POLAND

C++ GTEST GMOCK CODE REVIEW TDD

Worked in a scrum team implementing new 5G network critical features. Contributed new functionalities to the codebase and wrote comprehensive unit and component tests to ensure high code coverage. Participated in large-scale inter-team integration testing, requiring deep technical expertise and strong communication skills.

ML Engineer Student Worker, Milestone Systems A/S SEP 2023 – JUN 2025

1 YEAR 10 MONTHS • PART-TIME • COPENHAGEN, DENMARK

MACHINE LEARNING PYTHON LLM PROMPT ENGINEERING LEARNING AGILITY

Explored new technologies and solutions for video security systems in a fast-paced research team. Developed VLM RAG system, this included LLM Prompt Engineering, database integration, and providing frontend interface. Implemented real-time ML inference pipelines for video security systems using C++.

Master Thesis Student, GN Store Nord A/S FEB 2025 – JUN 2025

4 MONTHS • THESIS COLLABORATION • COPENHAGEN, DENMARK

MACHINE LEARNING PYTHON DSP RESEARCH EVALUATION

Evaluated various pitch-shifting and time-stretching algorithms as ML data augmentation methods for speech denosing. Assessed realism of the transformation and their implementation performance.

OTHER EXPERIENCE

Low-Latency Pitch-Shifting with STN, DAS | DAGA 2025 MAR 2025

TERM PAPER • PUBLICATION

Developed a low-latency pitch-shifting audio plugin based on fuzzy sines-transient-noise decomposition. Prototyped the solution in Python. Implemented as a real-time VST plug-in with C++ + JUCE framework. Published in the Conference Proceedings of DAS | DAGA 2025

Timbre Toys, Timbre Tools Hackathon 2024 MAR 2024

48 HOURS • FULL-TIME PARTICIPATION

Won the first prize in the Best Responsible Hack category. Implemented a timbre manipulation tool that allows the users to interact with a timbre the same way they would interact with a physical object. The main inspiration behind the project was a sonic interaction and a live performance.

Software for groundwater intakes, intakewise JUN 2021 - PRESENT

4+ YEARS • FREELANCE

ASP.NET MVC EF CORE SQL JAVASCRIPT CSS

Built a full-stack web app using ASP.NET MVC, which allows users to monitor a state of wells and water tanks, and alerts the users about sudden changes.

Mensa Poland FEB 2019 - PRESENT

in linkedin portfolio github

EDUCATION

MSc of Sound and Music Computing

2023-2025 • AALBORG UNIVERSITY COPENHAGEN

Gained in-depth knowledge of signal processing, physical modeling, audio perception, audio software and hardware design, development and evaluation. Studied in a Problem-Based-Learning system.

Master Thesis: Comparison of pitch-shifting and time-stretching methods for machine learning.

Bachelor of Computer Science

2018-2022 • AGH UNIVERSITY

Graduated from one of the most renowned Computer Science study programs in Poland, where I have learned about theoretical foundations and practical skills in building modern software and hardware.

Bachelor Thesis: A system supporting execution of hydrological modeling software in a computing cloud

TECH SKILLS

LANGUAGES C / C++ PYTHON C# / .NET
MATLAB JAVASCRIPT / TYPESCRIPT

DEVOPS DOCKER MULTI-OS DEVELOPMENT
LINUX AZURE GITHUB ACTIONS

OTHER DIGITAL SIGNAL PROCESSING
MACHINE LEARNING RESEARCH EVALUATION
LLM SQL FIGMA

WAYS OF WORKING GIT JIRA TDD SCRUM
AGILE CODE REVIEW

SOFT SKILLS

ATTENTION TO DETAIL PROBLEM SOLVING
TEAM PLAYER BUSINESS INTELLIGENCE
LEADERSHIP TIME MANAGEMENT
DELEGATION OF WORK PRESENTATION SKILLS
LEARNING AGILITY

LANGUAGES

- Polish – Native Proficiency
- English – Full Professional Proficiency
- Danish - Elementary Proficiency